

Structure of Finite Algebras

Given a field K , we define a K -algebra as a ring A with an (unital) ring inclusion $K \subset A$. This makes A into a K -vector space. We are interested in the case when A is a finite dimensional vector space over K . In such a case we say A is a finite K -algebra. (The term “finite dimensional” would be confusing since there is *also* a notion of the dimension of the ring A .)

Similar arguments to what we present below also apply to rings A which are set-theoretically finite. The reader is encouraged to work out the details.

We will restrict our attention to the commutative case. However, a similar structure result is valid for the non-commutative case.

The key observation is that any ideal in A is also a K -subspace of A .

Any chain of ideals in A is finite.

Moreover, each ideal is a finite dimensional K -vector space.

Every ideal in A is finitely generated.

(You can also use the previous statement to prove this statement!)

The Jacobson radical is nilpotent

The Jacobson radical J of a commutative ring is defined as the intersection of all its maximal ideals.

Exercise: For every $a \in J$ show that $1 + a$ is a unit in A . (Hint: $1 + a$ does not lie in *any* maximal ideal.)

Exercise: For every $a \in J$ show that there is a $b \in A$ and $n \in \mathbb{N}$ such that $a^n(1 - ab) = 0$. (Hint: Consider the decreasing sequence of ideals:

$$aA \supset a^2A \supset \cdots \supset a^nA \cdots$$

Combining the two exercises, we see that every $a \in J$ is nilpotent. Since J is finitely generated, it is nilpotent.

Co-primality

Given a maximal ideal M and a set $\{M_1, \dots, M_k\}$ of maximal ideals *different* from M , we can find, for each i an element $a_i \in M_i \setminus M$. From the primality of M it follows that $a_1 \cdots a_k$ cannot be in M . This gives an element in $(M_1 \cap \cdots \cap M_k) \setminus M$.

It follows that $M \cap (M_1 \cap \cdots \cap M_k)$ is *strictly* smaller than M .

Secondly, the image of this element in the field A/M is non-zero, hence invertible. It follows that there is a $b \in A$ such that $ba_1 \cdots a_k$ has image 1 in A/M and 0 in $A/(M_1 \cap \cdots \cap M_k)$.

Exercise: Show that the natural map

$$A \rightarrow \frac{A}{M} \times \frac{A}{M_1 \cap \cdots \cap M_k}$$

is surjective. (Hint: The element $e = ba_1 \cdots a_k$ gives an idempotent for the first factor and $1 - e$ gives an idempotent for the second factor.)

Chinese Remainder Theorem

Given distinct maximal ideals $M_1, \dots, M_k$, the above result shows that we have a *strictly* decreasing chain of ideals

$$M_1 \supset M_1 \cap M_2 \supset \cdots \supset M_1 \cap \cdots \cap M_k$$

The length of any such decreasing chain cannot be bigger than the dimension of A (since M is a proper subspace of A).

Thus we see that there are finitely many maximal ideals in A ; let these be denoted as $M_1, \dots, M_k$.

Exercise: Prove that the natural map

$$A \rightarrow \frac{A}{M_1} \times \cdots \times \frac{A}{M_k}$$

is surjective. (Hint: Use induction on k .)

The decomposition of A

Note that the kernel of the above homomorphism is J which is nilpotent. Hence, by Hensel lifting, the corresponding idempotents can be lifted to give a decomposition:

$$A \cong A_1 \times \cdots \times A_k$$

where the image of M_i is the only maximal ideal in A_i . It follows that $M_i^{n_i} = (0)$ in A_i for some n_i . Thus, this is the same as the decomposition

$$A \cong \frac{A}{M_1^{n_1}} \times \cdots \times \frac{A}{M_k^{n_k}}$$

As a corollary, we note that if A has no nilpotent elements, then A is a product of fields.